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1. Study of Various Network Commands used in Linux and Windows.

AIM: Study of Various Network Commands used in Linux and Windows.

Basic Networking Commands:

1) arp -a : Displays the IP and MAC addresses of devices in ARP table

2) hostname : Displays the name of your Computer.

3) ipconfig/all : Displays detailed TCP/IP Configuration, including IP, DNS, DHCP, and gateway details.

4) nbtstat -a : Displays NetBios information.

5) netstat : Displays active network connections and network statistics.

6) nslookup : Finds DNS information, such as the IP address of a domain.

7) ping : Tests network connectivity between two devices using ICMP.

8) Route : Displays or modifies the IP routing Table.

9) IP Command:

~ip address show

Purpose: Display IP address assigned to network interfaces.

10) MTR:

~mtr google.com

Purpose: Displays the route and network statistics for each hop.

11) TCP Dump:

- tcpdump -i eth0.

Purpose: Captures and displays network packets.

12) NMCLI:

~nmcli Connection show.

Purpose: Displays network Manager Connection profiles.

Result:

Thus, the Command to run network configuration, IP address, MAC address, routing info, DNS and Connection Status were displayed successfully.

Sample Output:

C:\> arp -a

Interface: 192.168.1.10

Internet Address	Physical Address	Type
192.168.1.1	00-11-22-33-44-55	dynamic

C:\> hostname

DESKTOP-PC

C:\> ipconfig /all

IPv4 address: 192.168.1.16

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

C:\> net stat -r

Network destination	Gateway
0.0.0.0	192.168.1.1

C:\> nslookup www.google.com

Name: www.google.com

Address: 142.250.195.68.

C:\> ping google.com

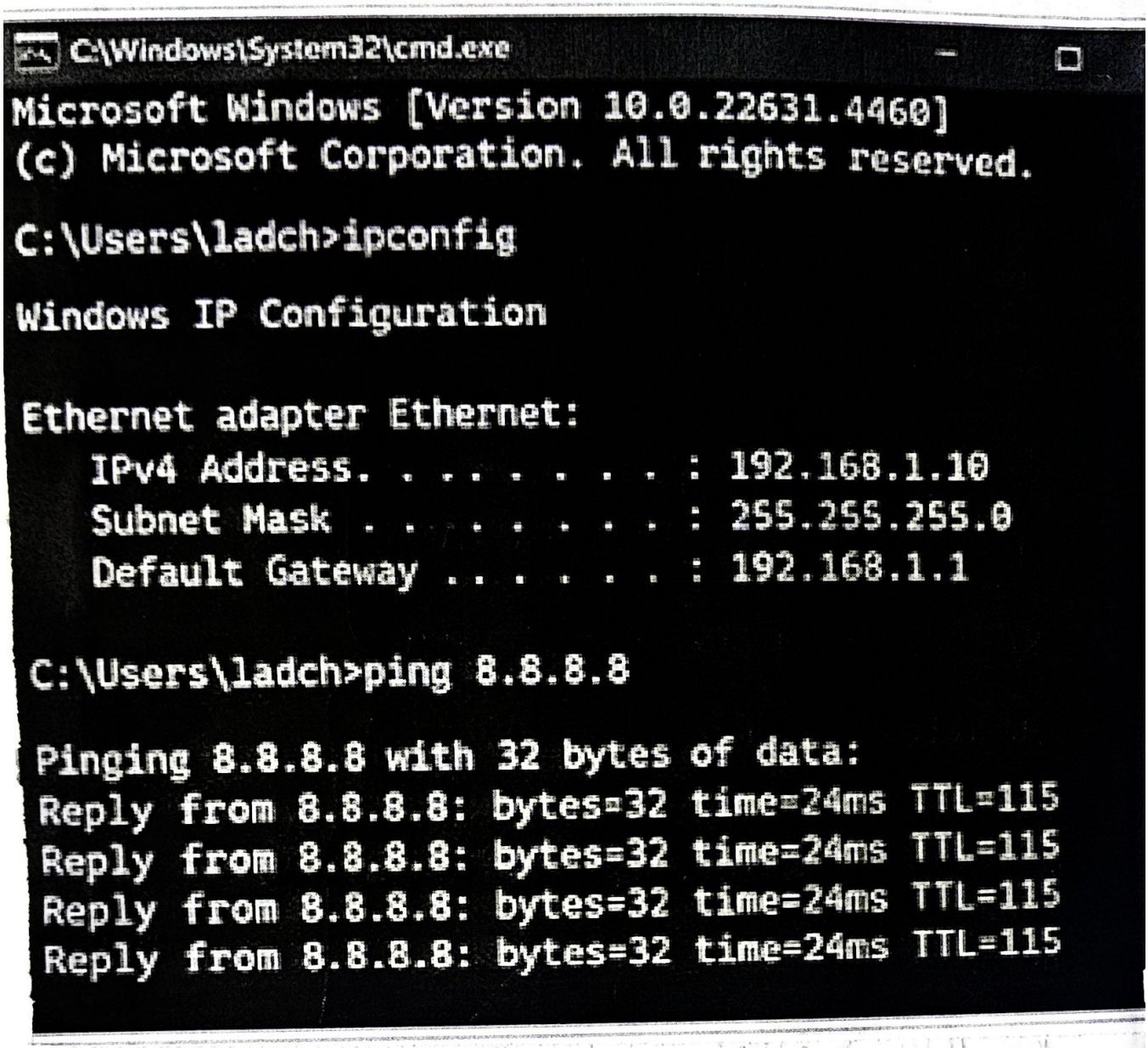
Reply from 142.250.195.68. bytes = 32 time = 20ms

TTL = 117.

Reply from 142.250.195.68 bytes = 32 time = 18ms,

TTL = 117.

1. Network Commands (Linux & Windows)



Ex no: 2
14/7

Networks Cables.

AIM:

To study different types of Network Cables and make an Ethernet Cross-over / Straight Cable

Types of Network Cables:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted pair (STP) Cable
3. SSTP Cable
4. Coaxial Cable
5. Fibre optic Cable.

Tools and parts required:

1. Ethernet Cable (CAT5 / CAT6)
2. Crimping tool.
3. Two RJ45 plugs.
4. Plug Shield (optional)

Procedure:

A) Study of Networks Cables

- Identify the difference types of Network Cables.
- Study their Categories, data transmission rates, advantages, disadvantages and applications.

B) Making ethernet Cross-Event / Straight Cable.

1. Thread the Shield onto the Cable.
2. Strip approximately 1.5 cm of Cable

Sheilding from both ends using the Crimping tools.

2. Untagle the four twisted pairs of wires.

4. Arrange the wires according to the required wiring arrangements.

5. Align the wires evenly and insert them into the RJ45 plug.

6. Ensure that the cable sheilding reaches the notch of the RJ45 plug.

7. Insert the RJ45 plug into the Crimping tool and press it carefully.

8. For a Crossover cable, use arrangement ~~A~~ one end and B at the other end.

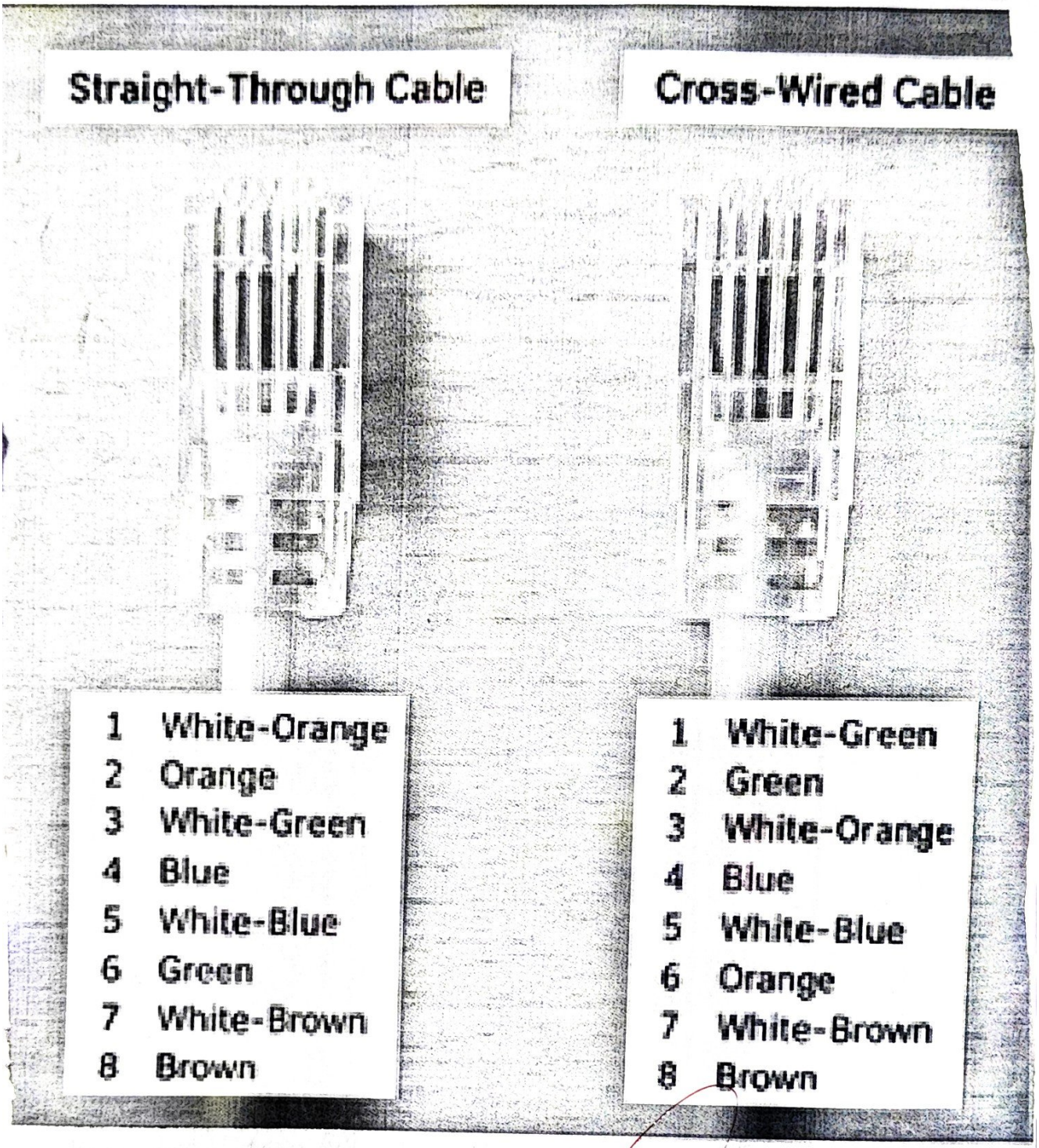
9. For a straight - through cable, use arrangement A at both ends.

10. Test the cable by connecting the required devices.

Result:

Thus, different types of network cables were studied and an Ethernet Cross-over / Straight cable was successfully made and tested.

2. Network Cables



Ex.No: 3

9/1/7

Study of Cisco Packet Tracer and Analysis of Network Devices.

Aim:

To study the installation and user interface of Cisco packets tracer and to understand the environment for designing a simple network devices such as hub and switch using Cisco packet Tracer.

Requirements:

*) Cisco Packet Tracer
*) 4 Generic PCs
*) 1 Hub
*) 1 Switch
*) Copper Straight-Through Cables.

Algorithm:

1. Open Cisco Packet Tracer.
2. Place 4 PCs and a Hub.
3. Connect all PCs to the Hub.
4. Assign IP addresses to the PCs.
5. Send a PDU from one PC to another.
6. Observer how the hub forwards the packet.
7. Create another network using 4 PCs and a Switch.
8. Assign IP addresses.
9. Send a PDU and observe the packet flow.
10. Compare hub and Switch.

Procedure:

Network using hub.

1. Open packet Tracer.
2. Place 4 PCs and 1 hub.
3. Connect all PCs to the hub using Copper Straight-Through Cable.
4. Configure the IP addresses.

PC	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

5. Select Add Simple PDU
6. Click PC0 and then PC1
7. Observe the packet flow.

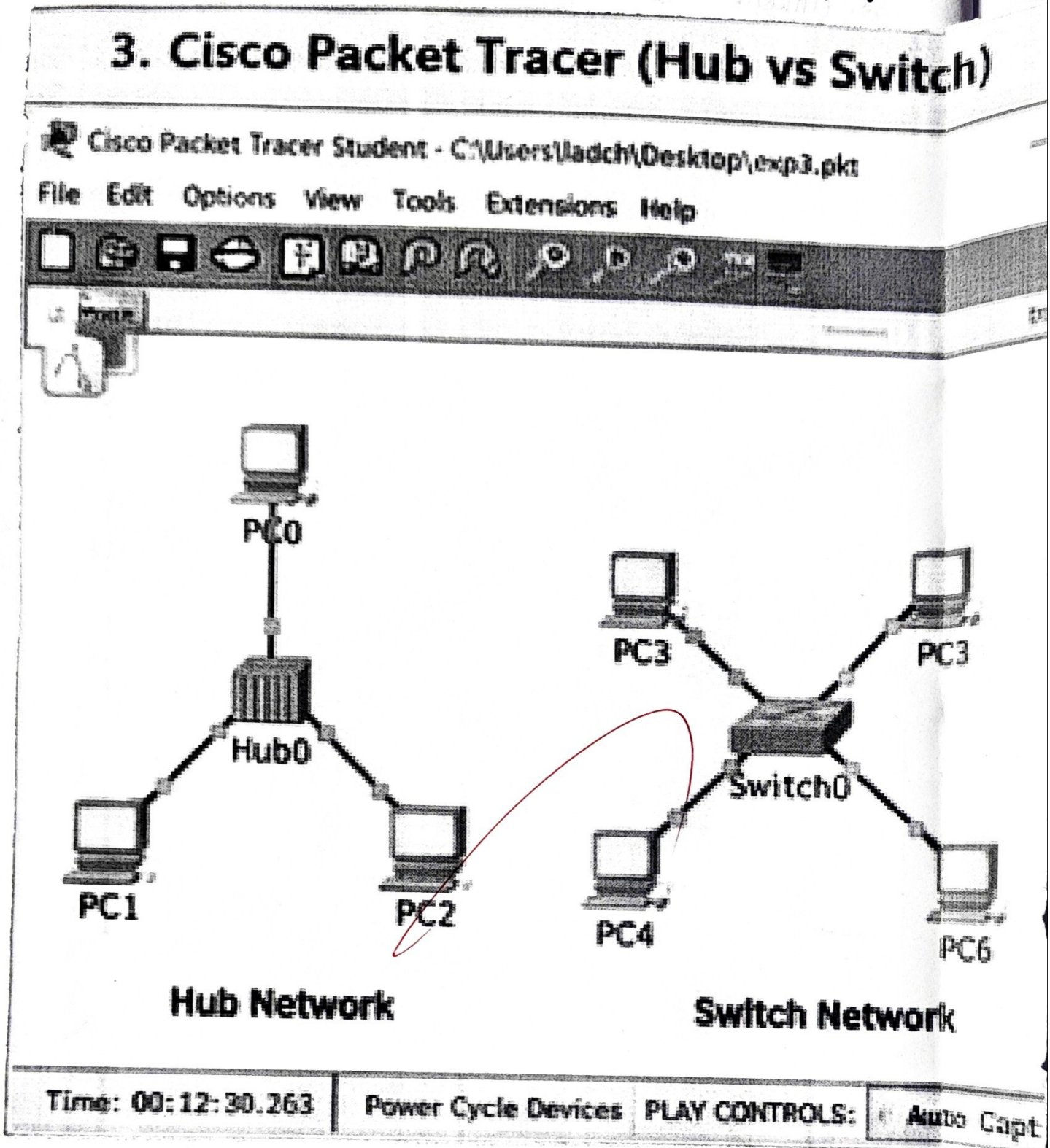
Network using Switch:

1. Place 4 PCs and 1 Switch
2. Connect all PCs to the Switch
3. Assign IP addresses.
4. Send a PDU from PC0 to PC1.
5. Observe the packet flow.

Result:

Thus, Cisco packet Tracer was studied successfully. The behavior of hub and Switch was observed and Compared.

3. Cisco Packet Tracer (Hub vs Switch)



Ex.No.4
2/1/7

LAN Setup Using Switch.

Aim:

To setup and configure a LAN using a switch and ethernet cables and check the connecting between computers.

Requirements:

- 4 PCs
- 1 Switch
- 4 Ethernet cables.

Algorithm:

1. Connect 4 PCs to a switch.
2. Assign IP addresses.
3. Check the connections using ping.
4. Share a folder between computers.
5. Check whether the shared folder can be accessed.

Procedure:

1. Connect four PCs to the switch using Ethernet cables.
2. Assign the IP addresses.
3. Open Command prompt.
4. Check the IP using:

ipconfig

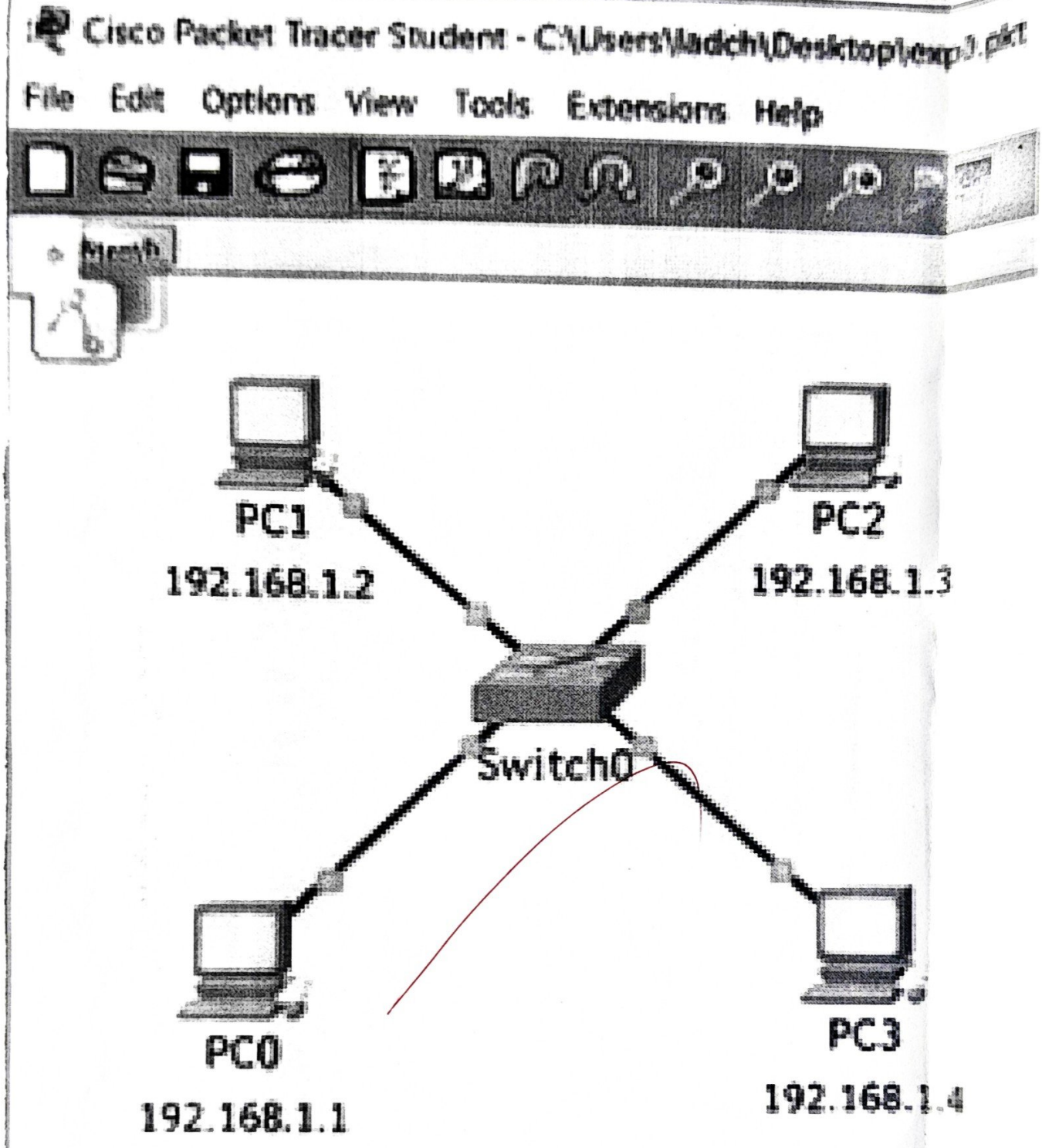
5. From PC1, check the connection to PC2 using:

Ping 10.1.1.2.

Result:

Thus, a LAN was successfully created and configured using a switch and Ethernet cables. Connecting between the computers was verified using the ping command.

4. LAN using Switch



Ex. No. 5
28/7

Wire Shark.

AIM:

To Study the Wire Shark packet capture and analyze packets at different layers of the protocol Stack.

Procedure:

1. Open Wire Shark.
2. Select the local Area Connection / Ethernet Interface.
3. Start packet Capture
4. Capture approx 100 packets.
5. Stop the Capture
6. Apply filters for
   - → TCP / UDP
   - → ARP
   - → DNS
   - → HTTP
   - → IP / ICMP
   - → DHCP
7. Select individual packets
8. Observe the packet list, packet details and packet Bytes panes
9. Analyze the protocol fields.
10. Save the Captured packets.

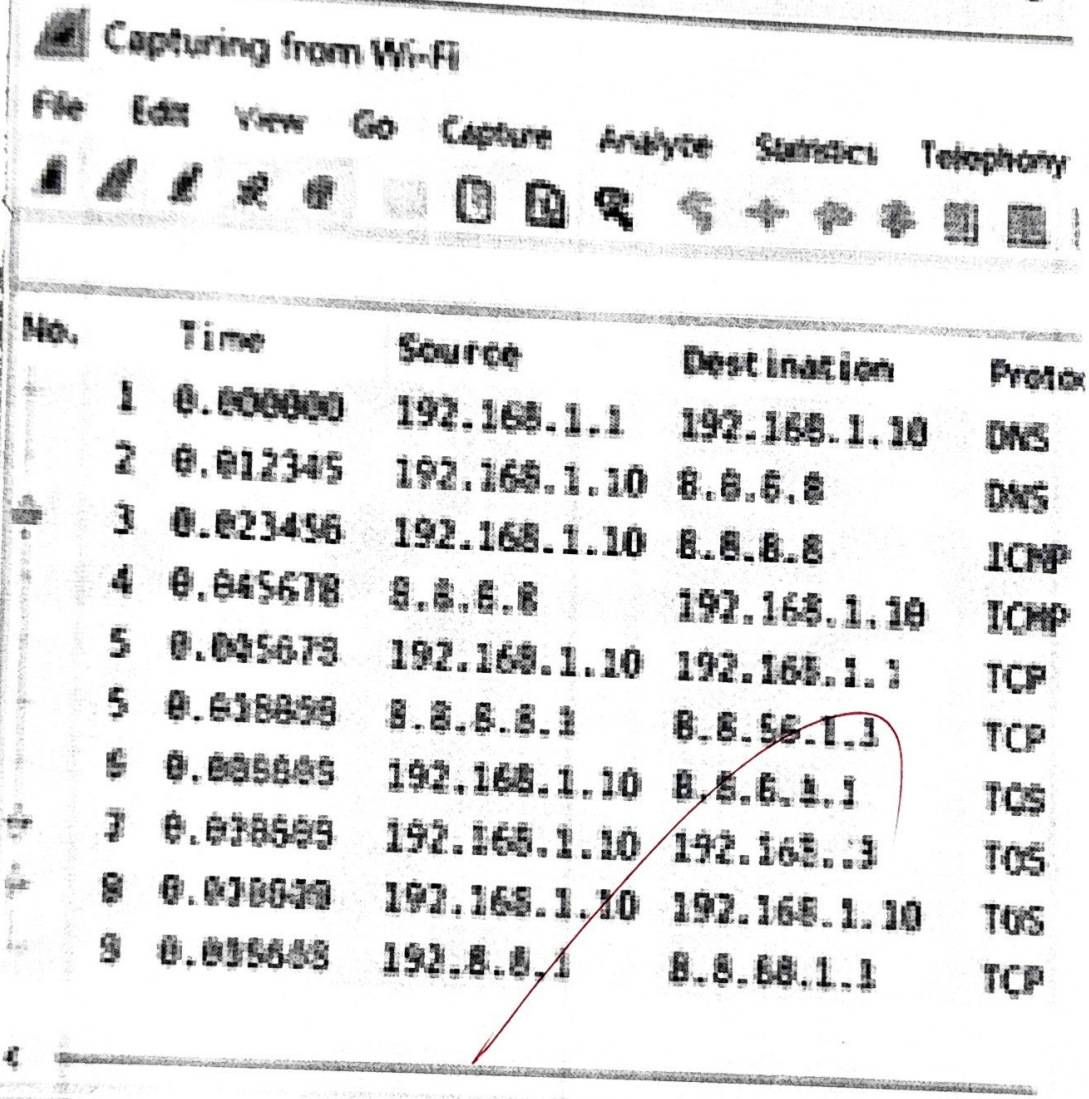
Result:

Thus wireshark was successfully used to Capture filters and Analyse network packets and their protocol fields.

SAMPLE Output:

No	Protocol	Source	Destination
1	ARP	192.168.1.10	Broadcast
2	DNS	192.168.1.10	8.8.8.8
3	TCP	192.168.10.10	142.x.x.xx.xx
4.	HTTP	192.168.1.10	Server.

5. Wireshark Packet Cap



Ex. No. 6
4/8

Hamming Code.

AIM:

To Implement error detection and Correction using hamming Code and Verify Correction Capability using a test data stream.

Procedure:

Sender:

1. Create a Sender program.
2. Read a text message from the input field.
3. Convert the text into binary.
4. Calculate the insert the required redendant parity bits.
5. Generate the hamming Code.
6. Store the encoded data in a file named Channel.

Receiver:

7. Read the encoded data from Chanel.
8. Recalculate the parity bits.
9. Determine whether an error is present.
10. If an error is detected, find its position
11. Correct the erroneous bits.
12. Remove the redundant bits.
13. Convert the Corrected binary data back into ASCII text.

16. Display the recovered message.

Result:

Thus the hamming code was successfully implemented to detect and correct a single bit error in the transmitted data.

Sample Output:

Enter the text: HELLO

Binary Data:

01001000 01000101 01001100 01001100 01001111

Hamming Code:

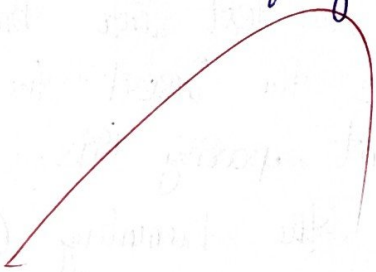
- - - -

Error detected at position : 12

Error corrected Successfully

Decoded Data:

HELLO.



6. Hamming Code (Error Detection & Cor



IDLE Shell 3.10.6

File Edit Shell Debug Options Window H

```
Enter 4-bit data : 1011
Hamming Code :  0110011
Received Code :  0110001
Error detected at position :  3
Corrected Code :  0110011
Corrected Data :  1011
>>>
```

Ex.no 7
11/8

Sliding Window Protocal.

AIM:

To implement flow control at the data link layer using sliding window Protocol and simulate the transfer of frames between two nodes.

Procedure:

Sender:

1. Get the window size from the user.
2. Read a text message.
3. Consider one character as one frame.
4. Assign a frame number to each character.
5. Create frames in the format.
6. Store the frames in sender_Buffer.
7. Send frames according to the window size.
8. Wait for acknowledgement from the reciever.
9. Check the ACK number.
10. If the ACK is correct, transmit the next set of frames.
11. If NACK is recieved, retransmitted the required frames.

Reciever:

12. Read frames from sender-Buffer.

7. Sliding Window Protocol (Flow Control)

```
Enter window size : 3
Total number of frames : 7

Sending frames...
Frame 0 sent        ACK received
Frame 1 sent        ACK received
Frame 2 sent        ACK lost  ->  Retransmitting
Frame 2 sent        ACK received
Frame 3 sent        ACK received
Frame 4 sent        ACK received
Frame 5 sent        ACK received
Frame 6 sent        ACK received

Transmission completed successfully.
```

100.8
8/8

NMAP LIVE HOST Discovery.

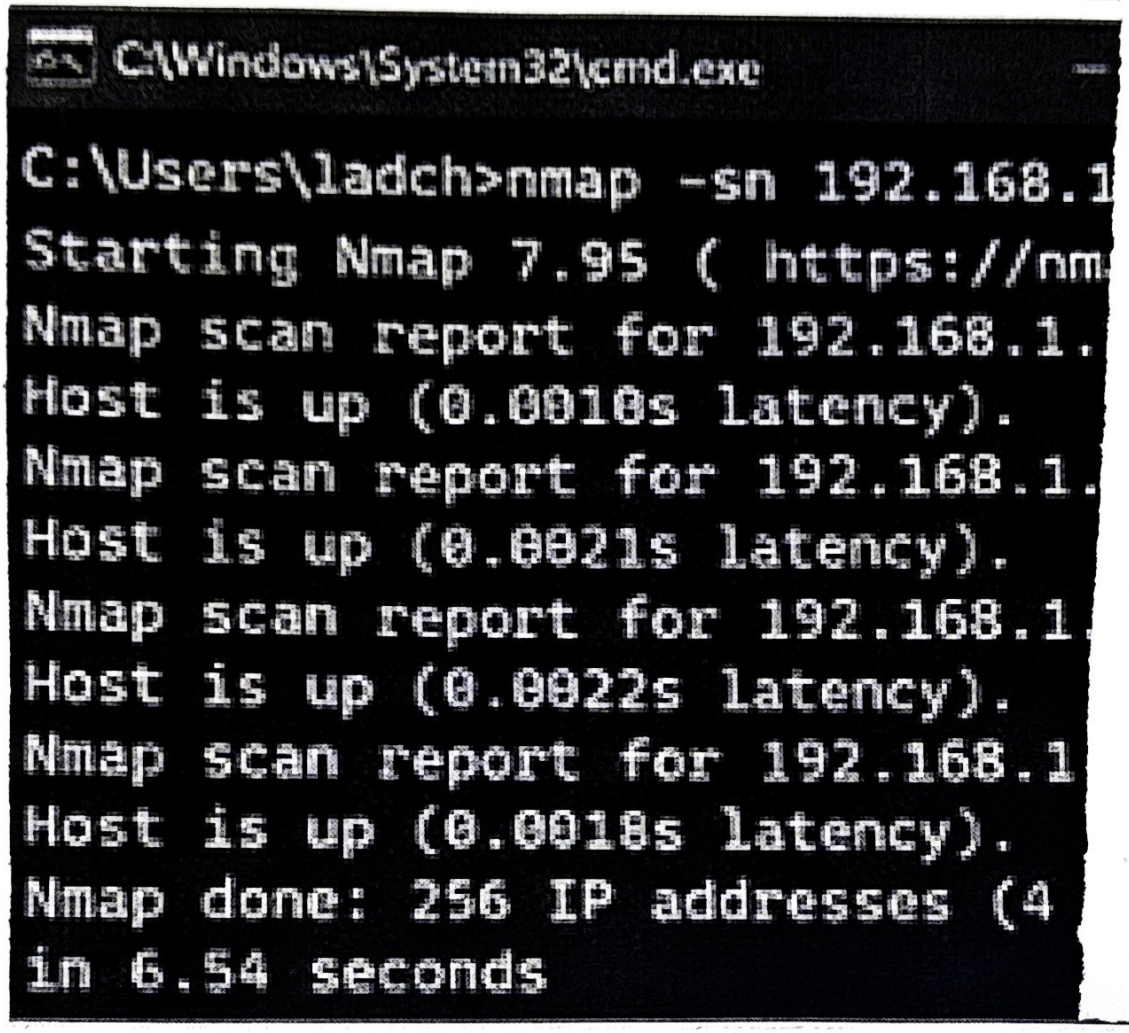
AIM:

To descover live host using NMAP scans based on ARP, ICMP, TCP and UDP on the Try HackMe NMAP platform.

Procedure:

1. Open Nmap room on TryHackMe
2. Study the target network and subnet structure.
3. Identify the target IP addresses.
4. Perform ARP - based host discovery.
5. use:
   nmap -PR -sn Target.
6. nmap -PE -sn TARGET
7. nmap -PP -sn TARGET
8. nmap -PS 23 TARGET.
9. Perform TCP ACK ping using the appropriate -PA option.
10. Observes the live hosts identified by Nmap.
11. Study reverse DNS lookup using -R.

8. Nmap (Live Host Discovery)



Result: Thus, NMap was successfully used to discover live hosts using ARP, ICMP, TCP and UDP based techniques.

9
3

Subnetting using Cisco Packet tracker.

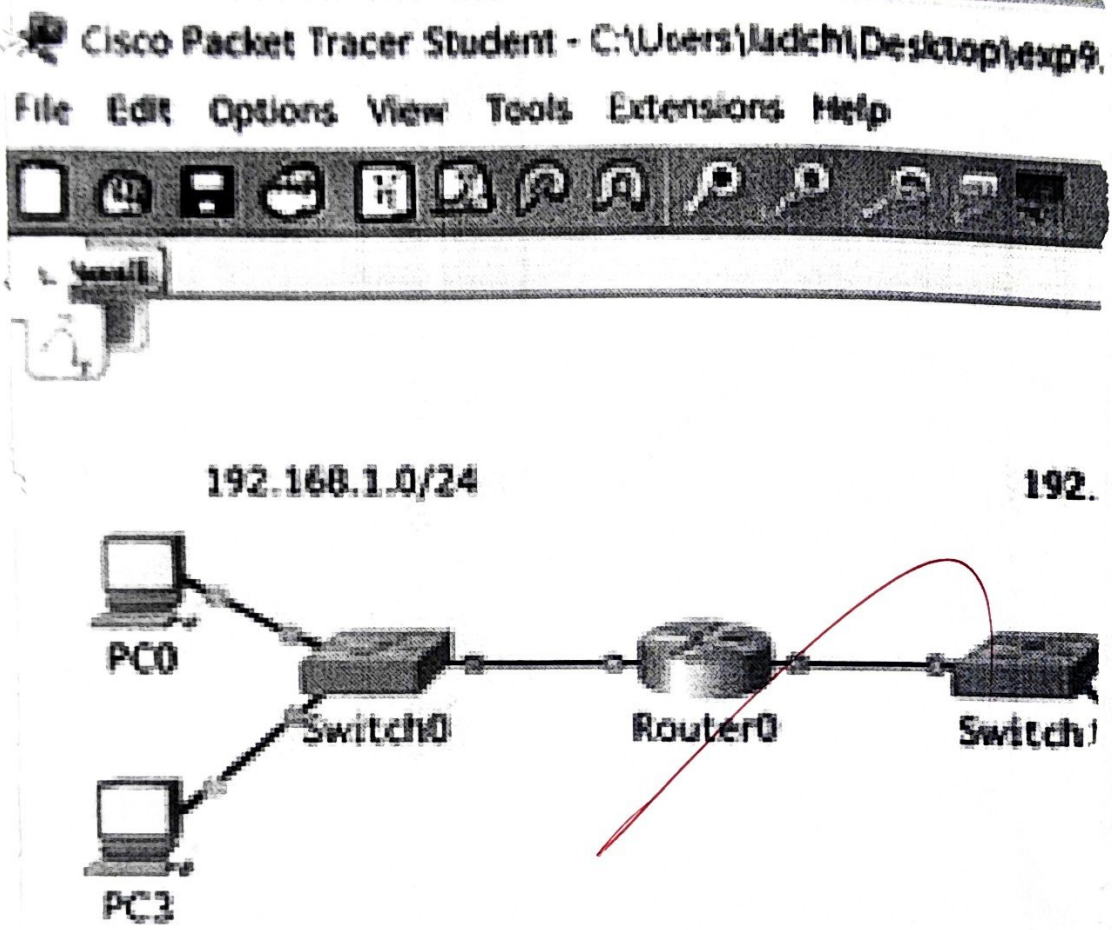
AIM:

To implement classes IP subnetting using Cisco packet Tracker and establish communication between multiple subnets trough routes.

Procedure:

1. Open Cisco packet tracer.
2. Create a new network topology.
3. Add routers, switches and PCs.
4. Connect the devices using appropriate cables.
5. Divide the network 192.168.1.01.24 into Smaller Subnets.
6. Use 127 subnet masks as specified in the manual.
7. Configure the router interface with Suitable IP address.
8. Configure the switch ports.
9. Enable router interfaces using no Shut down.
10. Test connectivity using ping
11. Observe packet transmission btw Subnets.

9. Subnetting in Packet Trace



Interneticoorking from Routers.

Aim:

To design and Configure an Internework using routers in Cisco Packet tracker and Verify Communication between different networks.

Procedure:

1. Open Cisco Packet tractor.
2. Place one router and two PCs.
3. Connect the PCs to the router using Copper Straight through cables.
4. Configure Router().
5. Enter privilege mode : enable.
6. Enter Configuration mode; configt.
7. Configure Fast Ethernet 0/0
   interface Fast Ethernet 0/0.
   ip address 192.168.10.1.225.255.255.0
   no Shut down.
8. Configure PC0:
   IP address: 192.168.10.2
   Subnet Mask: 255.255.255.0
   Gateway: 192.168.10.1
9. Configure PC1
   IP address: 192.168.20.2
   Subnet Mask: 255.255.255.0
   Gateway: 192.168.20.1

10. Send a PDU from PC0 to PC1
11. Observe the packet transmission and acknowledgement.

Result:

Thus, an internetwork using a router was successfully designed and configured, and communication between networks was verified.

Sample Output:

Router # show ip route

R 192.168.20.0/24 [120/1]

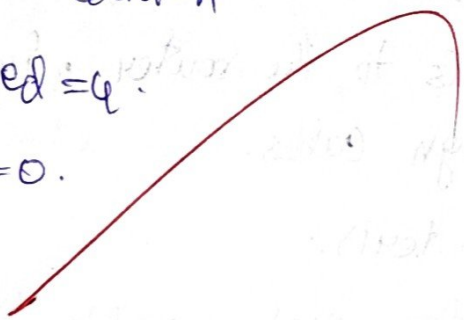
via 192.168.10.2.

Ping Successfully.

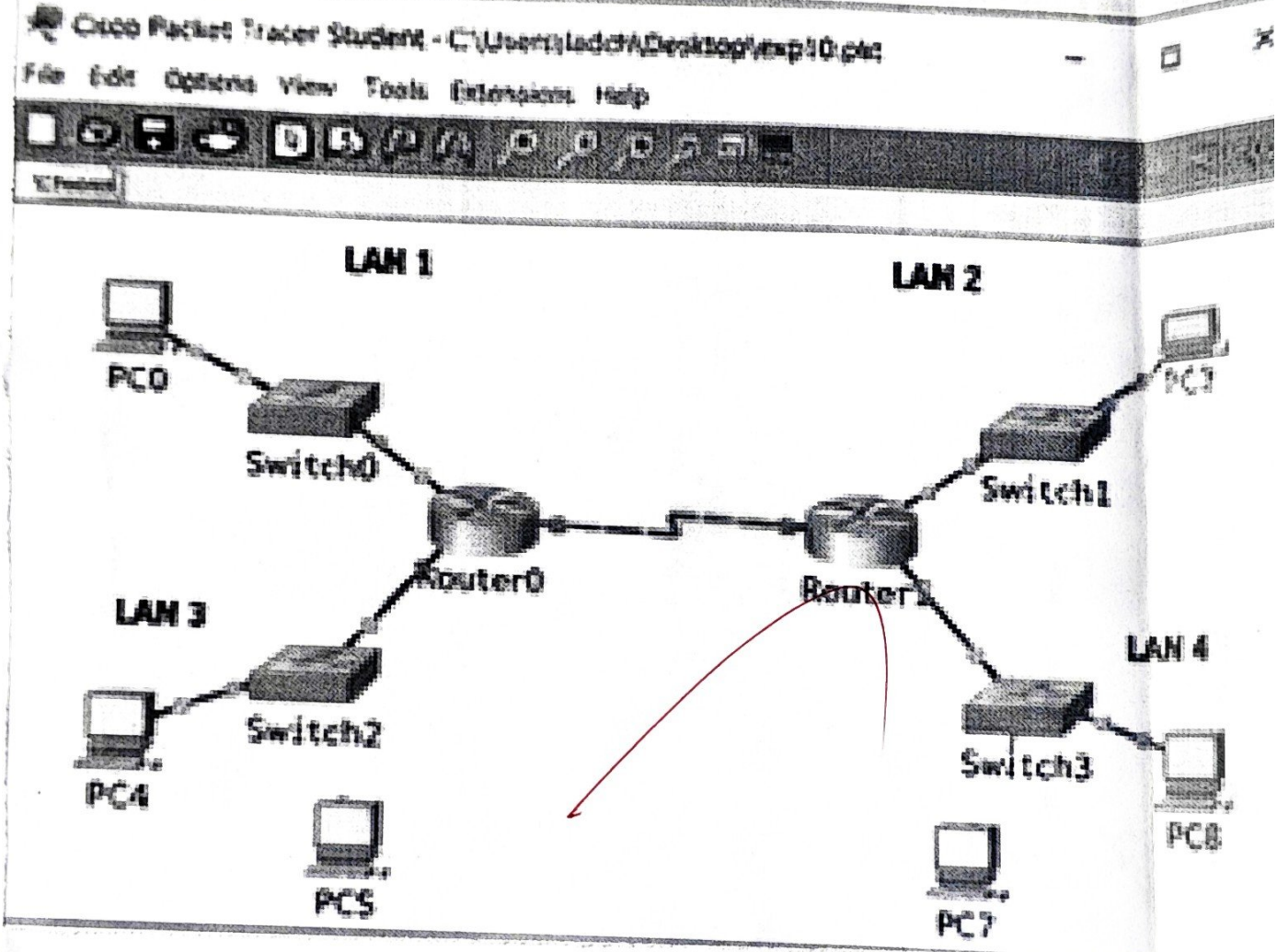
Packets: Sent=4

Received=4.

Lost=0.



10. Internetworking with Routers



Ex.no.11
8/9

Static Routing and RIP

AIM:

To Simulate Static Routing and RIP using Cisco Packet Tracer.

Procedure:

1. Open Cisco packet tracer.
2. Place two routers and Suitable PCs.
3. Connect the routers and PCs.
4. Assign IP addresses to all interfaces.
5. Configure routers interfaces using ip address no Shutdown.
6. Add Static router using.

   Ip route < destination- networks
   <Subnet. mask > <next- hop>

7. Verify the routing table using:

   Show ip route.

8. Test Connectivity using ping.

RIP:

9. Configure the router interfaces.
10. Enter RIP configuration mode:

    router rip.
    Version 2.

11. Add the Connected network.

network <network-address>.

12. Repeat the configuration on the other router.

13. Verify the routing table.

14. Ping a host in the remote networks.

Result.

Thus, Static Routing and RIP was successfully configured and the connectivity btw different networks was verified.

11. Routing at Network Layer (Static

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Router(config)#ip route 192.168.2.0 255.255.255
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router#show ip route
Codes: C - connected, S - static, R - RIP

C       192.168.1.0/24 is directly connected, Gi
C       10.0.0.0/30 is directly connected, Gigi
S       192.168.2.0/24 [1/0] via 10.0.0.2
R       192.168.3.0/24 [120/1] via 10.0.0.2
Router#
```

TCP/UDP Socket Communication

AIM:

To implement end-to-end communication at the transport layer using TCP/UDP sockets and develop a simple chat application.

Procedure:

Echo client - Server

1. Create a Server program.
2. Create a client program.
3. Create a socket at the Server.
4. Bind the socket to an IP address.
5. Make the Server listen for client connections.
6. Accept the client connection.
7. The clients sends a message.
8. The Server receive and echoes the message.

Chat Program:

10. Create client and Server Socket programs.
11. Establish a connection.
12. Read messages from the user
13. Send the messages through the Socket.
14. Continue communication until the user terminates the connection

Result:

Thus, end-to end communication was successfully implemented using socket programming and echo/chat communication between client and server was verified.

Sample output:

Server:

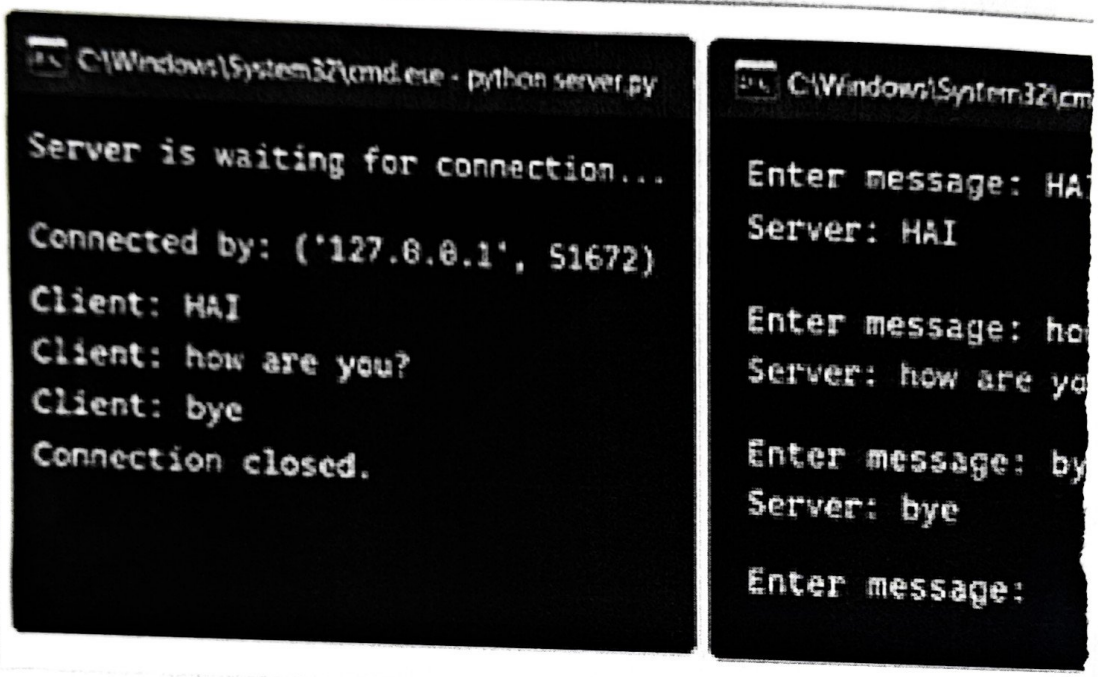
Server started...
Waiting for client.
Client Connected.

Received: Hello Server.
Echo: Hello Server.

Client:

Connected to Server.
Enter message: Hello Server
Server replied:
Hello Server.

12. End-End Communication (TCP/UDP So



13

Own PING Program.

AIM:

To implement a simple ping program using Socket programming and verify the reachability of a destination host.

Procedure:

1. Create a program to accept a destination IP address / hostname.
2. Creates a socket for Network Communication.
3. Generate an ICMP echo Request.
4. Send the request to the destination
5. Wait for an ICMP echo reply.
6. Calculate the response time.
7. Display whether the host is reachable.
8. Repeat the process for multiple requests
9. Display the process transmission statistics.

Result:

Thus, a simple ping program was successfully implemented and reachability of the destination host was verified.

Sample Output

Enter destination: 8.8.8.8

PING 8.8.8.8

Reply from 8.8.8.8

Sequence=1 Time = 94ms.

Reply from 8.8.8.8

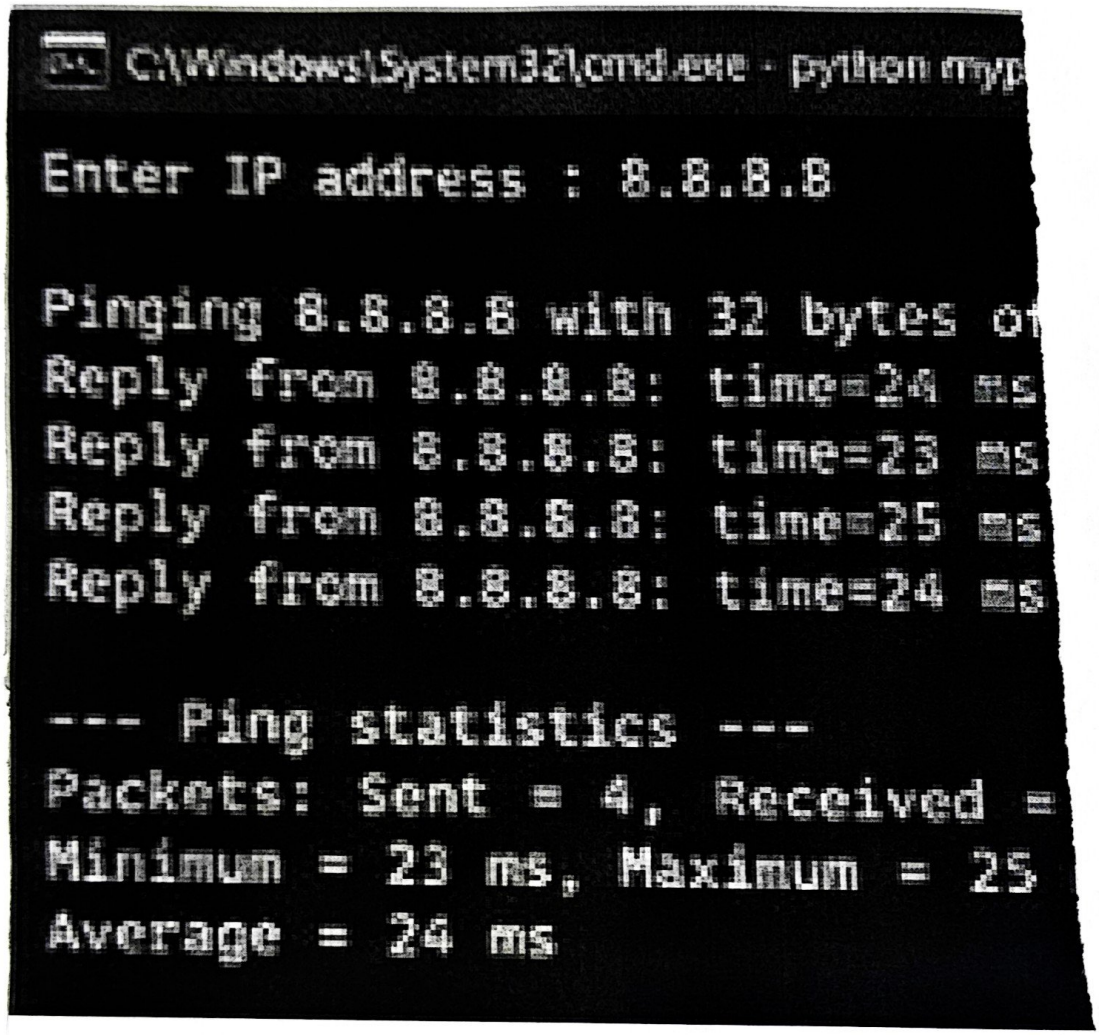
Sequence = 3 Time = 25ms.

Packets sent : 2

Packets Received : 3.

Packets loss: 0%

13. Own Ping Program



Ex.No.13
22/9

Aim:
To impl
Socket p
a destina

Procedure:
1. C
2.
3.
4

Name
28/9

RAW Socket Packet Sniffing

AIM:

To implement packet sniffing using RAW sockets and analyze the packets received through the network interface.

Procedure:

1. Create a RAW socket using the Socket programming interface.
2. Select the required network interface.
3. Enable packets reception through the RAW socket.
4. Capture incoming network packets.
5. Read packet header.
6. Extract important fields such as.
   - Source IP
   - Destination IP
   - Protocol
   - Packet length.
4. Identify the protocol of the captured packet.

8. Display the packet information.
9. Continue capturing packets until the program is terminated.

Result:

Thus, RAW sockets were successfully used to capture and analyse network packets.

Sample output:

RAW Socket Packet Sniffer.

Packet Captured.

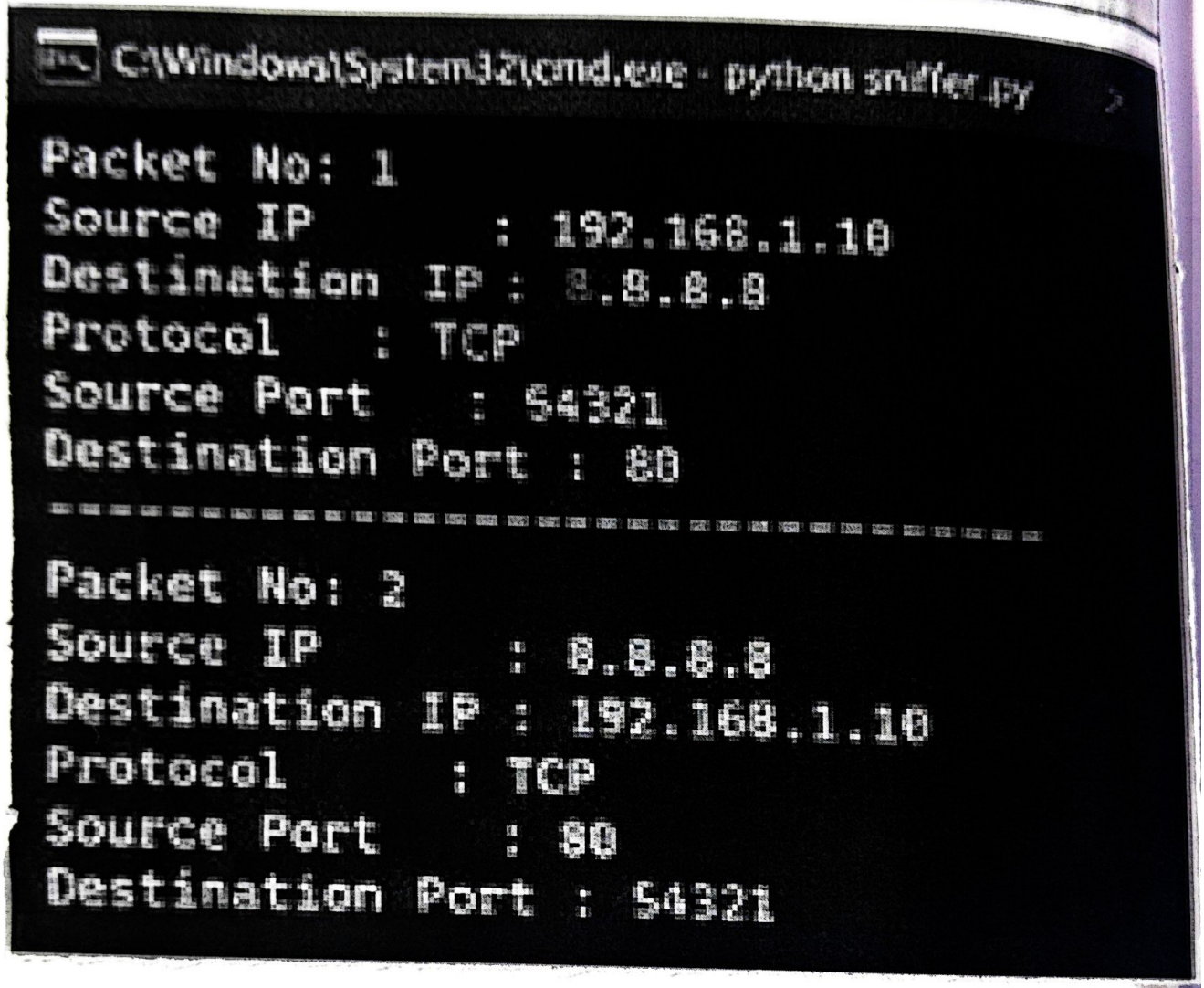
Source IP: 192.168.1.10

Destination IP: 192.168.1.1

Protocol: TCP

Packet length: 60 bytes.

14. RAW Socket Packet Sniffing



Completed

13/10/26